Modular Arithmetic, Residues & Power Cycles

Core router: TARGET MODULUS -> REDUCE STATE -> LEGAL OPERATIONS -> CYCLE / COMBINE -> CHECK

Congruence means same remainder, not ordinary equality. The symbol `a congruent b (mod m)` means `m divides a-b`. This divisibility fact is retrieved from NT-01; the modular notation and legal operations are developed here.

1. Meaning and legality

Safe operations: add, subtract, multiply, and raise congruent quantities to positive integer powers. Reduce residues early.

Cancellation is conditional. From `ac congruent bc (mod m)`, cancel c modulo m only when gcd(c,m)=1. Example: `2*1 congruent 2*4 (mod 6)`, but 1 is not congruent to 4 mod 6.

If `3x congruent 6 (mod 7)`, inverse of 3 is 5, so `x congruent 2 (mod 7)`.

2. Power cycles and last digits

Modulo 7, powers of 2 cycle `2,4,1`. Therefore `2^100 mod 7 = 2`. Last digit means mod 10; last two digits means mod 100.

For `5^k mod 100`, exponent 2 already gives 25 and multiplying by 5 returns 25. Hence every exponent k>=2 ends in 25.

For `n^n mod 7`, a universal period must preserve both the base modulo 7 and, for nonzero base states, the exponent modulo 6. The source anchor period is 42.

3. Simultaneous congruences

`x congruent 2 (mod 3)` and `x congruent 3 (mod 5)` combine to `x congruent 8 (mod 15)`. With shared-factor moduli, check compatibility first: residue differences must agree modulo the gcd of the moduli.

Contrast: equality vs congruence; divisibility vs congruence; brute-force powers vs cycles; legal vs illegal cancellation; compatible vs incompatible simultaneous congruences.

4. Attempt before help

A. Find 3^100 mod 7. H3: list powers until 1 returns. H2: identify the cycle. H1: huge power -> finite residue state. H0: last digit of 7^2026.

B. Solve 4x congruent 3 (mod 7). H3: inverse of 4 is 2. H2: check gcd(4,7)=1. H1: division requires invertibility. H0: analyze 6x congruent 9 (mod 15).

C. Solve x congruent 1 (mod 4), x congruent 4 (mod 5). H3: x=1+4k and test mod5. H2: parametrize one class. H1: combine repeating lists. H0: x congruent 2 (mod6), x congruent5 (mod9).

Recognition & First-Line Lab

1. Show 38 congruent 3 (mod 5) using divisibility.
2. Decide whether 17=5 and whether 17 congruent 5 (mod12).
3. Reduce 12345 modulo7 before further work.
4. Write first four residues of 2^n mod7.
5. Last digit of 3^100: state the modulus.
6. Last two digits of 7^50: state the modulus.
7. From 3x congruent6 (mod7), write the inverse step.
8. From 2x congruent4 (mod6), state why direct cancellation is unsafe.
9. First line: x congruent2 (mod3), x congruent1 (mod4).
10. Compatibility: x congruent1 (mod4), x congruent2 (mod6).
11. Distinct residues A,B modulo m: write collision condition to avoid.
12. For n^n mod7, name the two periodic pieces.
13. Explain congruence without congruence notation.
14. Find inverse candidate for5 modulo12.
15. Does6 have inverse mod15?
16. If a congruent b mod9, state a valid relation for squares.

Practice Ladder - selected transfer

1. Find 2^40 mod7.
2. Find last digit of 7^2026.
3. Find last two digits of 5^37.
4. Solve 3x congruent5 (mod7).
5. Solve x congruent2 (mod3), x congruent3 (mod5).
6. Analyze 4x congruent8 (mod12) and list all classes mod12.
7. Determine compatibility of x congruent1 (mod6), x congruent4 (mod9).
8. Find least positive x: x congruent3 (mod4), x congruent5 (mod7).
9. Give a counterexample to unrestricted cancellation modulo8.
10. Solve 6x congruent9 (mod15).
11. Find last two digits of 11^2026.
12. Find smallest m>=14 for which 1^4,...,14^4 have distinct residues.
13. Explain why a universal period of n^n mod7 must be divisible by7 and6.
14. WHY-NOT: cancel 10 from 10x congruent20 (mod30)?
15. Clock state t congruent2 (mod4), t congruent5 (mod6): does it exist?

H0 Mixed Mastery

1. 83 congruent11 (mod12): write the divisibility statement.
2. Find first line for 7^123 mod10.
3. 4x congruent5 (mod9): write the legality check.
4. x congruent2 (mod6), x congruent5 (mod9): write compatibility check.
5. Find 2^2026 mod7.
6. Find last two digits of 5^2026.
7. Solve 5x congruent7 (mod12).
8. Solve 6x congruent12 (mod18) as classes mod18.
9. Least positive x: x congruent1 (mod4), x congruent4 (mod5).
10. Find 3^1234 mod7.
11. Does x congruent2 (mod4), x congruent3 (mod6) have a solution?
12. Find last digit of 7^(7^7).
13. Compare 14|42 with 42 congruent0 (mod14).
14. Compare cancellation in 3x congruent6 mod7 and mod9; solve both.
15. Find smallest m>=14 with distinct residues of 1^4,...,14^4.
16. A learner expands 2^100 to find mod7. Give the shorter route and why it works.

Source anchors

IOQM-2024-Q03 -> 25; IOQM-2024-Q23 -> 31; IOQM-2025-Q20 -> 42. Exact historical wording remains source-controlled.